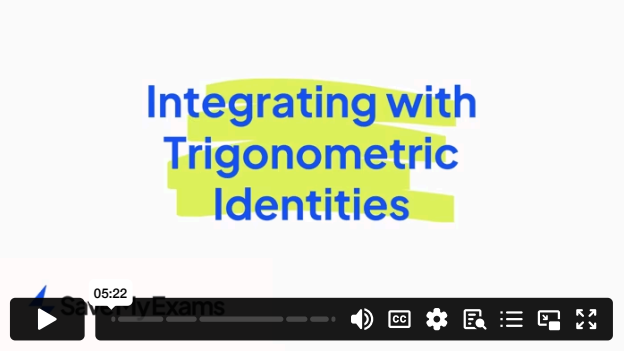
Integrating with Trigonometric Identities

Exam code: 9MA0

Download PDF

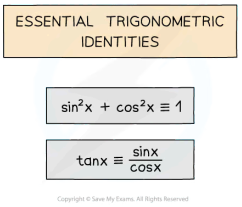
Written by: Paul
Updated on 16 February 2026



Did this video help you? Yes No

Integrating with trigonometric identities

What are trigonometric identities?



- Some are given in the formulae booklet
  - Be sure to note the difference between the ± and ∓ symbols!

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B} \qquad \left(A \pm B \neq \left(k + \frac{1}{2}\right)\pi\right)$$

$$\sin A + \sin B = 2\sin\frac{A+B}{2}\cos\frac{A-B}{2}$$

$$\sin A - \sin B = 2\cos\frac{A+B}{2}\sin\frac{A-B}{2}$$

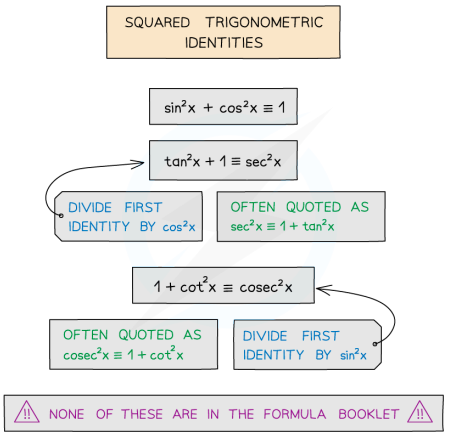
$$\cos A + \cos B = 2\cos\frac{A+B}{2}\cos\frac{A-B}{2}$$

$$\cos A - \cos B = -2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$$

Copyright © Save My Exams. All Rights Reserved

How do I know which trigonometric identities to use?

- There is no set method
- This is a matter of experience, familiarity and recognition
  - Practice as many questions as possible
  - Be familiar with trigonometric functions that can be integrated easily
  - Be familiar with common identities – especially **squared** terms
  - **$\sin^2 x$**, **$\cos^2 x$**, **$\tan^2 x$**, **$\text{cosec}^2 x$**, **$\sec^2 x$**, **$\cot^2 x$** all appear in identities



Copyright © Save My Exams. All Rights Reserved

How do I integrate $\tan^2 x$, $\cot^2 x$, $\sec^2 x$ and $\text{cosec}^2 x$?

- The **integral** of **$\sec^2 x$** is **$\tan x$ (+c)**
  - This is because the **derivative** of $\tan x$ is $\sec^2 x$
- The **integral** of **$\text{cosec}^2 x$** is **$-\cot x$ (+c)**
  - This is because the **derivative** of $\cot x$ is $-\text{cosec}^2 x$
- The **integral** of **$\tan^2 x$** can be found by using the identity to rewrite $\tan^2 x$ before integrating:
  - $1 + \tan^2 x = \sec^2 x$
- The **integral** of **$\cot^2 x$** can be found by using the identity to rewrite $\cot^2 x$ before integrating:
  - $1 + \cot^2 x = \text{cosec}^2 x$

How do I integrate expressions involving sin x and cos x?

2A

- **sin 2A = 2sinAcosA**

e.g. $\int \sin x \cos x \, dx$

$\sin 2A = 2\sin A \cos A$

$= \int \frac{1}{2} \sin 2x \, dx$

$= -\frac{1}{2} \times \frac{1}{2} \int -2\sin 2x \, dx$

"ADJUST" AND "COMPENSATE"

$= -\frac{1}{4} \cos 2x + c$

AN EQUIVALENT ANSWER CAN BE FOUND USING THE FOLLOWING TECHNIQUE

Copyright © Save My Exams. All Rights Reserved

- **$\sin^n kx \cos kx$** or **$\sin kx \cos^n kx$** can be integrated using **reverse chain rule** or **substitution**
- Notice no identity is used here but it looks as though there should be!

e.g. $\int \sin x \cos^3 x \, dx$

$= -\frac{1}{4} \int -4\sin x \cos^3 x \, dx$

REVERSE CHAIN RULE

"ADJUST" AND "COMPENSATE"

$-\sin x$ IS DERIVATIVE OF $\cos x$

$= -\frac{1}{4} \cos^4 x + c$

NOTICE NO IDENTITY USED IN THIS QUESTION

Copyright © Save My Exams. All Rights Reserved

- $\sin^2 kx$ and $\cos^2 kx$ can be integrated by using the identity for cos 2A
  - For $\sin^2 A$, $\cos 2A = 1 - 2\sin^2 A$
  - For $\cos^2 A$, $\cos 2A = 2\cos^2 A - 1$

e.g. $\int \sin^2 x \, dx$

$\cos 2A = 1 - 2\sin^2 A$

SO $\sin^2 A = \frac{1}{2}(1 - \cos 2A)$

$= \int \frac{1}{2}(1 - \cos 2x) dx$

$= \int \left(\frac{1}{2} - \frac{1}{2}\cos 2x\right) dx$

$= \int \left[\frac{1}{2} - \frac{1}{2} \times \frac{1}{2}(2\cos 2x)\right] dx$

$= \frac{1}{2}x - \frac{1}{4}\sin 2x + c$

Copyright © Save My Exams. All Rights Reserved

How do I integrate tan x?

FORMULA BOOKLET

Copyright © Save My Exams. All Rights Reserved

- This is a standard result from the formulae booklet

How do I integrate other expressions involving trig functions?

- The formulae booklet lists many standard trigonometric derivatives and integrals
  - Check both the "Differentiation" and "Integration" sections
  - For integration using the "Differentiation" formulae, remember that the integral of **f'(x)** is **f(x)** !

DIFFERENTIATION

$f(x)$	$f'(x)$
$\tan kx$	$k\sec^2 kx$
$\sec kx$	$k\sec kx \tan kx$
$\cot kx$	$-k\,\mathrm{cosec}^2 kx$
$\mathrm{cosec}\, kx$	$-k\,\mathrm{cosec}\, kx \cot kx$

INTEGRATION (+ CONSTANT)

$f(x)$	$\int f(x)dx$
$\sec^2 kx$	$\frac{1}{k}\tan kx$
$\tan kx$	$\frac{1}{k}\ln\lvert\sec kx\rvert$
$\cot kx$	$\frac{1}{k}\ln\lvert\sin kx\rvert$
$\mathrm{cosec}\, kx$	$-\frac{1}{k}\ln\lvert\mathrm{cosec}\, kx + \cot kx\rvert$, $\frac{1}{k}\ln\left\lvert\tan\left(\frac{1}{2}kx\right)\right\rvert$
$\sec kx$	$\frac{1}{k}\ln\lvert\sec kx + \tan kx\rvert$, $\frac{1}{k}\ln\left\lvert\tan\left(\frac{1}{2}kx + \frac{1}{4}\pi\right)\right\rvert$

Copyright © Save My Exams. All Rights Reserved

- Experience, familiarity and recognition are important – practice, practice, practice!
- Problem-solving techniques

- MIXTURE OF x's AND 2x's
- $\sin^2 x$ IS GOOD–COMMON IDENTITY
- $1 + \cos 2x$ IS NOT A GOOD DENOMINATOR

$$= \int \left(\frac{2\sin^2 x}{1 + (2\cos^2 x - 1)} + 1 \right) dx$$

$$\boxed{\cos 2A \equiv 2\cos^2 A - 1}$$

$$= \int \left(\frac{2\sin^2 x}{2\cos^2 x} + 1 \right) dx$$

$$= \int (\tan^2 x + 1) dx$$

$$= \int \sec^2 x \, dx$$

$$\boxed{\sec^2 x \equiv 1 + \tan^2 x}$$

$$= \tan x + c$$

A STANDARD RESULT — BUT UNDER DIFFERENTIATION IN THE FORMULAE BOOKLET

Copyright © Save My Exams. All Rights Reserved

Examiner Tips and Tricks

- Make sure you have a copy of the formulae booklet during revision.
- Questions are likely to be split into (at least) two parts:
  - The first part may be to show or prove an identity
  - The second part may be the integration
- If you cannot do the first part, use a given result to attempt the second part.

Worked Example

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them

Test yourself

Was this revision note helpful? Yes No

Previous Next

Course Study tools All topics

Resources

Learning Hub
Scholarship
Join
Past Papers
Solution Banks
Sitemap

Dashboard

Overview
Brand ambassador
Profile
Subscription
Settings
Log out

Company

About us
Exam Specificity
Content Quality
Promotions
Jobs
Terms
Privacy
Cookie Policy
Help and Support

Subjects

Biology
Chemistry
Physics
Maths
Geography
English Literature
Psychology
All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.